



Reliability of Electronic Evidence: an application for model-based auditing

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ABSTRACT

Much legal evidence is being generated by and stored in information systems. In this paper we look at evidence from an auditing point of view. Auditors rely on evidence of the party being audited, who may have a legitimate or illegitimate interest to manipulate it. To assess the quality of audit evidence, we argue for an approach called *model-based auditing*. It is based on a mathematically precise model of the expected relationships between the flow of money and the flow of goods or services. Such equations are used for cross verification. If the equations do not hold, either something is wrong (violation) or some underlying assumption is false (exception). To show the usefulness of the approach, we look in particular at a case study of a legal dispute about automated contract monitoring. A precise revenue model is instrumental in demonstrating that the data set does indeed constitute appropriate evidence to settle the case.

Keywords

auditing; electronic evidence

1. INTRODUCTION

There has been a lot of interest in the nature of legal evidence, see for example Wigmore [39] or Anderson et al [2]. The debate concerns admissibility of evidence, strength of evidence, and the interpretation of sources of evidence to establish facts that are relevant to some legal dispute. Increasingly, also stories [34] and argumentation are used to combine raw facts into a legally meaningful account of events [3]. The typical example concerns criminal law: do the available sources of evidence establish that the defendant is guilty? Here the evaluation usually takes place *after* the events, for instance in court. Instead we will discuss what constitutes legal evidence in a corporate setting of establishing compliance, or contract delivery. Settling such questions can also be meaningful *before* the events have taken place, as part of the design of procedures, processes and information systems.

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Much legal evidence is being generated by processes and systems which are under control of the party being regulated. To ensure reliability of such evidence, certain provisions must be built in. These provisions are called internal controls [11]. Regulators will want to audit the adequacy and operating effectiveness of such control measures. For example, as part of prudential supervision, financial regulators regularly audit the processes of banks to establish reliability and compliance with financial regulations. Consider for example the Basel-III rules, which require banks to send regular reports about the amount of capital they hold in various risk categories. Such reports are meaningless, if the numbers cannot be relied on.

The audit process involves interpretation and debate [8]. Therefore audits are an instance of what Black calls regulatory conversations [5]: they help establish common understanding about what constitutes compliance in a certain domain. For instance, Power [28] argues that a pervasive 'logic of auditability' has come to dominate thinking about risk management, which lies at the heart of much corporate regulation. This alleged influence warrants a study of evidence from the point of view of auditing theory. That suggests the following research question:

How should auditors assess reliability of evidence that is being generated by information systems?

The realization that compliance is being designed, is well accepted in Business Process Management, e.g. [31]. Governatori and colleagues call this shift towards preventative measures to ensure the compliance of business processes 'compliance by design' [15]. However, the business process community typically uses BPMN or Petri nets to formally represent operational processes. They reduce compliance to properties that are essentially about the order, nature and presence of activities. Instead, we focus on the financial results. We will therefore take economic transactions to be the central notion. Note that many inter-organizational business processes are concerned with establishing a transaction, see for instance work on value modelling [14, 12].

In order to establish the quality of evidence concerning a series of transactions, we need a precise mathematical model of the expected relationships between the flow of money and the flow of goods or services in a transaction (value-cycle), expressed as a series of audit equations. Given a model, hypotheses can be generated for testing, and requirements about evidence can be formulated. This computational approach to auditing is called *model-based auditing* [10, 37], by analogy with approaches like model-based diagnosis [30],

which are also based on a precise model of the domain, rather than on heuristics. Value-cycle models can be derived from practices in the accounting domain [33].

In this paper we investigate the usefulness of model-based auditing for assessing the reliability of legal evidence that is being generated by information systems.

To express the value-cycle models and audit equations and to execute the tests, we use a dedicated programming language developed by Paul Griffioen, called Pacioli [16]. An important feature of Pacioli is that it allows transactions to be expressed in their precise units of measurements. So a shipment of shoes is counted in pairs, not in dollars, as this may be subject to valuation differences.

To give a ‘proof of concept’ and show that such a model-based perspective on the generation of evidence makes sense, we study a case about automated monitoring of contract delivery. The case concerns a legal dispute over a contract for public transport. We reconstruct the reasoning of an expert about the quality of evidence, in particular about how it was generated. The legal interpretation process which proceeds from ‘raw facts’ (steps in a business process) to ‘events’ (commitment, execution) and finally to ‘conditions’ that have meaning in terms of the contract (delivery, payment), are crucial. This should make the case relevant, even though it does not directly concern regulatory compliance.

The paper is constructed as follows. In Section 2 we provide some background on evidence in auditing theory. In Section 3 we discuss reasoning about evidence in AI and law. In Section 4 we present our specific approach, model-based auditing and the Pacioli programming language. Section 5 discusses their application in the case study.

2. AUDITING

Companies collect evidence of behaviour and produce reports, so they can be held accountable. To verify these reports auditors or inspectors are called in. “Auditing is the systematic process of objectively obtaining and evaluating evidence regarding assertions about economic activities and events to ascertain the degree of correspondence between the assertions and established criteria, and communicate the results to interested users” [1]. In the typical audit engagement, we identify three actors: management, stakeholders, and auditors (Figure 1). Management is accountable for its actions to the owners or shareholders of a company, who want to see a return on investment. It is also accountable to other members of society, like employees, suppliers or others who are dependent on the company. In case of regulatory compliance, the stakeholders (citizens) are represented by a government regulator. Management prepares statements about some object, for instance about accuracy and completeness of financial results, reliability of a computer system, or compliance of a process. Auditors verify these statements, and provide assurance as to their reliability, based on evidence. The collection of evidence, the test of controls and the substantive testing all take place according to standards. In accounting, standards like the Generally Accepted Accounting Principles (GAAP) are well established, but in other fields, like IT audit or regulatory compliance, the standards are often written as open norms. Open norms are general principles or objectives; they still have to be made specific for the situation in which they are applied [23, 38]. That means that many decisions are left to the ‘professional judgement’ of the auditor. Auditors are ex-

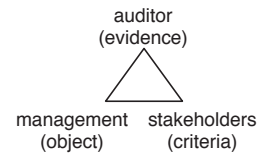


Figure 1: Audit engagement

pected to provide “reasonable assurance that the (financial) reports are free from material misstatements” [22]. Unlike what is required in legal testimony, certainty ‘beyond reasonable doubt’ cannot be provided, as the auditor cannot be present all the time and must rely on accounts from the party being audited, so there is always a chance of manipulation or error. A misstatement or error is called material, when it would affect any auditor’s judgement. In practice, materiality is often interpreted as a threshold on the size of an error, to affect judgement.

The current ISA 500 standard issued by the International Federation of Accountants (IFAC) mentions evidence as follows: “The objective of the auditor is to design and perform audit procedures in such a way as to enable the auditor to obtain sufficient appropriate audit evidence to be able to draw reasonable conclusions on which to base the auditor’s opinion” [19][article 4]. There are two crucial notions: *sufficient* and *appropriate*.

Sufficiency refers to the quantity of audit evidence. This is used in the context of substantive testing, where auditors typically take samples to represent the behaviour of an entire population. There are formulas for calculating the minimally required size of a sample, given the size of the population, the required confidence level or detection risk, and the materiality or tolerance for error [18]. Well known methods are random sampling and monetary unit sampling, where the population is made up of the individual Euros or Dollars involved in the items to be tested (e.g. invoices), so that items involving large sums of money have a higher chance of being selected.

Appropriateness refers to the quality of audit evidence. It involves both *relevance* and *reliability*. “Relevance deals with the logical connection with, or bearing upon, the purpose of the audit procedure and, where appropriate, the assertion under consideration” [19][A27]. For example, the data collected must be about events in the period being audited; or it must apply to the aspects that were agreed on in the scope of the audit engagement. That also means that collecting more audit evidence may in general not compensate for its poor quality. In accounting practice, the concept of reliability breaks down into *accuracy* or *correctness* (statements correspond to reality), and *completeness* (all relevant aspects of reality are represented). Many also mention *timeliness*, but that can also be seen as part of accuracy. According to an influential standard (FASB 1980), “Reliability represents the extent to which information is unbiased, free from error, and representationally faithful” [25]. How are these aspects being evaluated? Potential bias depends on the source of the evidence. Presence of errors depends on consistency and the quality of internal control measures, discussed below. Finally, representational faithfulness involves accuracy and completeness, but seen as a property of the financial report as a whole: does it paint a ‘fair’ picture?

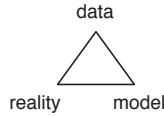


Figure 2: Model-based approach

mentation is also applicable to establish the appropriateness of evidence for a purpose, as required by [19].

Third, *narratives* are based on psychological accounts of sense-making and memory [34]. Sources of evidence are interpreted in a way that makes it possible to combine them into a believable story. The story that best explains the facts is then taken to be true. In this strict sense, narratives play no role in auditing. However, as one reviewer suggested, narratives represent a causal chain of events or scripts of how things are supposed to happen. In that sense they correspond to procedures or business processes. Verifying adherence to processes and procedures is crucial for auditing. In addition, as will become clear in Section 4 there is a strong link to value cycle models and the concept of tours. A tour is a completed run of a value cycle model [13].

In all these approaches the reasoning steps, sources of evidence, hypotheses and background knowledge have to be made explicit, so that sources of doubt can be identified [2]. One way to do so is to use a model. How does having a model affect the reasoning about audit evidence? For one thing, having a model can help make the essential distinction between an exception and a violation. This distinction is crucial for example in the field of deontic logic [26]. What is crucial to realize here, is that the model is normative: it prescribes the ‘ideal’ case in which people would actually follow the norm. The model generates expectations about behaviour, or hypotheses, which are then compared with data. The data are supposed to be a faithful representation of reality. So essentially we are dealing with a three-way relation (Figure 3). If a difference is found, it can be attributed to the model (in three ways), to the data, or to the situation itself. That suggests five possibilities.

- a. The model is not applicable (wrong model)
- b. The model is wrongly applied (wrong prediction)
- c. An exception is not covered by the model (model is wrong)
- d. The data do not faithfully represent reality (wrong data)
- e. There is a violation (situation is wrong)

In the case study, we also expect to find examples of these five types of counterarguments.

4. MODEL-BASED AUDITING

Model-based auditing is a computational approach to compliance monitoring and auditing, which is based on a normative meta-model of the value-cycle: the movement of money and goods or services [13, 17, 37, 10]. Where do these value-cycle models come from? In accounting practice, expertise can be found about the expected reporting risks of various business types. An example is the business typology developed over several decades by Starreveld and by Frielink, which used to be textbook material for Dutch auditors [33]. We refer to [7] for an exposition in English.

Audit Equations. Given a value-cycle model, hypotheses can be generated for cross-verification. For every main type of business (e.g. trade, manufacturing, services, etc.), the expected relationships between indicators representing the flow of money and the flow of goods or services are known. These relationships generate audit equations. For example: $revenues = quantity\ of\ goods\ sold * price\ of\ goods$, where price is fixed. Quantity of goods sold can be derived from the outflow from inventory, which is independently registered. A scenario of under-reporting revenues would be found out, by checking revenues against inventory movement. If segregation of duties are being enforced, and if other internal controls are in place, we expect reliable data from independent sources. Under such assumptions no transactions can be performed without leaving a trace. But if we cannot rely on segregation of duties, some transactions can be performed without leaving a trace. For example, if the inventory manager and the financial controller collude, under-reporting of revenues would not always be detected [13].

Actor Analysis. Based on the considerations listed above, and the discussion at the end of Section 3 we developed the following practical approach to actor analysis. These steps have been used successfully in teaching a postgraduate course on model-based auditing (Hulstijn p.c.)

1. Identify the *actors* involved and their general interests. Do some actors have countervailing interests?
2. For the main stakeholder, identify the *type of business*. Sketch a coarse business model: how are revenues being generated? Usually, it is about money against goods or services. This will determine the main value streams.
3. Identify the *key performance indicators* of these value streams (KPI). Sometimes you have to use ‘quasi goods’ (tickets), or side conditions (energy consumption) to make an intangible service measurable. What is their *unit of measurement*? For services, also take planning and forecasting into account. Now write down the main *audit equations*, with the corresponding units of measurement.
4. Determine for each term in these audit equations, whether actors have a legitimate or illegitimate reason to overstate the value (+), or rather understate its value (–), or even leave it out all together. This will determine whether you need a positive verification (accuracy), or a negative verification (completeness). Simplify the number of terms needed, by substitution.
5. Look at the business process. What are reliable data sources (messages; applications; databases) to measure these KPIs? How are they aggregated? Are the sources independent, or from actors with countervailing interests? For those KPIs for which you do not have data, try to reduce them to others, using the equations.

Value nets. The value cycle models from practice [33], can be formalized as a *value net*, illustrated in Figure 3. The notation is based on Petri nets, so events are depicted by rectangles and states are modelled by circles, or places, as they are called. Each event represents a transfer of values, just like in other formal approaches to value modelling, like e3-value [14], or DEMO [12]. In the examples we show here, the events are buy and sell transactions, but in other cases like manufacturing, transformations can also take place.

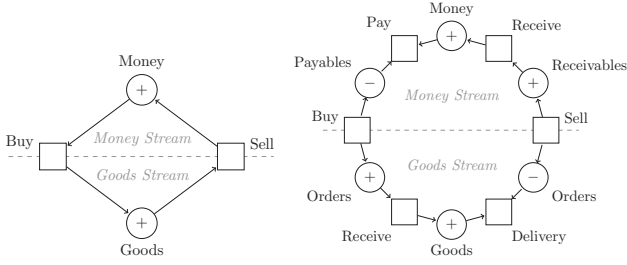


Figure 3: Value Nets for trading companies

We define a value net informally as a Petri net with some special characteristics. Value nets are like the audit nets introduced by Elsas [13], but they omit audit specific features like recordings and roles of agents. For our purposes, value nets are good enough.

DEFINITION 1. A *value net* is a Petri Net with the following characteristics

- The net is *cyclic* in the following sense. There is a special place labelled ‘money’ from which it is possible to reach every other node in the model and that can be reached by every node in the model. So every node in the model is on a path from ‘money’ to ‘money’.
- The net is divided in a money part and a production part, and ‘money’ is in the money part. Those transitions in which the input places are in the money part and the output places in the production part, or vice versa, are called *exchanges*. All other transitions are called *conversions*.
- Each place is labelled as *positive* (+) for assets, or *negative* (−) for liabilities.

Units of measurement. For the computation of normative behaviour, it is essential to record events in their proper unit of measurement. Units can be standard SI (metre, litre, etc.), specific (20 foot container) or combinations. For example, as we will see in the case, transport services are expressed in ‘occupied vehicle hours’. By contrast, in accounting often the flow of products or services is expressed in Dollars or Euros. This measure is commensurable, convenient and standard, but in this way important information is lost. Expressing inventory or sales in monetary units involves a valuation: how much are these assets worth? Valuation depends on the context (time and place), because of market conditions or fluctuations in currency rates. From an inventory expressed in pairs of shoes, it is always possible to calculate the value, but not conversely.

We use vector algebra and matrices to make the formalization scalable. Note that a weighted Petri Net can always be expressed by an incidence matrix M , where a cell m_{pt} contains a weight $n > 0$ when there is a link from place p to transition t . A dimensioned Petri Net associates units of measurement with each place.

DEFINITION 2. A *dimensioned Petri Net* is a tuple $(P, T, \mathbf{F}, \mathbf{B}, \mathbf{u})$ with P a finite collection of places, T a finite collection of transitions, $\mathbf{F}$ and $\mathbf{B}$ each an incidence matrix of the form $\mathbb{N}^{P \times T}$, and $\mathbf{u} \in \mathbb{U}^P$ a vector of units.

Here $\mathbf{F}$ refers to forward links (consume) and $\mathbf{B}$ to backward links (produce). Unit vector $\mathbf{u}$ couples a specific unit to each

place. A dimensioned marking is a pair $\mathbf{mu}$ from $\mathbb{N}^P \times \mathbb{U}^P$. For convenience, flow matrix $\mathbf{G}$ is defined by $\mathbf{G} = \mathbf{B} - \mathbf{F}$. The effect of a transition vector $\mathbf{t}$ on matrix $\mathbf{G}$ is given by the product $\mathbf{G} \cdot \mathbf{t}$.

Example. Consider the left hand diagram in Figure 3. Suppose the company is buying shipments of 20 pairs of shoes, for \$19 a pair, and selling them for €25 a pair. This value net is represented by the following two matrices.

Consume: $\mathbf{Fu}$			Produce: $\mathbf{Bu}$		
P \ T	Buy	Sell	P \ T	Buy	Sell
Money	380 \$		Money		25 €
Goods		1 pair	Goods	20 pair	

A valuation is a linear property of a marking with a monetary unit of measurement. It gives the total economic value of the tokens in a dimensioned net. A weight vector $\mathbf{v}$ contains the value of a token for each place. Given a marking $\mathbf{m}$ the inner product $\langle \mathbf{v}, \mathbf{m} \rangle$ is the total value.

DEFINITION 3. A valuation $\mathbf{v}$ of unit a is a proper valuation if $\mathbf{v}[p] = c \cdot a / \mathbf{u}[i]$ for each $p \in P_{\text{mon}}$ with conversion factor c such that $\mathbf{u}[p] = c \cdot a$.

For example if the valuation unit is Euro (€) and $\mathbf{u}[p]$ is dollar (\$) then $\mathbf{v}[p] = 1.12 \$/\text{€}$. This handles the units of measurement properly and maintains conservation of value.

Tours. Next we introduce the notion of a *tour*. A tour relates the traces of behaviour of an enterprise to the cyclic structure of its value net. The cyclic structure is not directly apparent in behaviour because transitions can fire at any point in the cycle. In a tour, we group those transactions that make up what Ijiri calls a causal chain of events [20]. Generally such a chain of events is a variation on buying products and resources, producing an end product, and selling the product. The cycle starts with the consumption of money and ends with the production of money.

A formal version of the tour concept was introduced in [13]. It defines a tour as a constellation of events whose total effect is expressed on money only. All other produced tokens are at some point consumed by another step.

DEFINITION 4. A transition vector $\mathbf{t}$ with firing result $\mathbf{m} = \mathbf{G} \cdot \mathbf{t}$ is a tour when $\mathbf{m}[\text{money}] > 0$ and $\mathbf{m}[x] = 0$ for any place $x \neq \text{money}$.

In this case, vector $\mathbf{t}$ is the number of times a specific event has to occur per tour. Every time a tour has occurred the amount of money increases by $\mathbf{m}[\text{money}]$. This increase is called the *value jump*. This formal tour concept therefore captures the essence of the notion of value. In the example above, a tour consists of a buy event, for 20 pairs at 380\$, combined with 20 individual sell events, each for 25€. If valued at 1.12 \$/€, the value jump is 180\$ for 20 pairs.

Factoring Tours. Tours consist of parts. In auditing we need to analyse various aspects of transactions. Therefore it makes sense to factor out or partition the space of possible behaviours. In Table 2 in the following section we show an example of a factored tour for the transport case.

We can reveal the various parts of a tour by factoring tour result $\mathbf{G} \cdot \mathbf{t}$. For brevity’s sake we ignore the minor technical issue of the place sign. Let partitioning $P = P_{\text{mon}} \cup P_{\text{prod}}$ denote the places in the money part and the production part. From the partitioning of the places also the transitions can be partitioned. For any transition $t \in T$, notation $\bullet t$ refers to origin of the transition, and $t^\bullet$ to the destination.

DEFINITION 5. Transitions are partitioned by equations

$$T_{\text{mon}} = \{t \in T \mid \bullet t \subseteq P_{\text{mon}} \wedge t^\bullet \subseteq P_{\text{mon}}\} \quad (1)$$

$$T_{\text{prod}} = \{t \in T \mid \bullet t \subseteq P_{\text{prod}} \wedge t^\bullet \subseteq P_{\text{prod}}\} \quad (2)$$

$$T_{\text{buy}} = \{t \in T \mid \bullet t \subseteq P_{\text{mon}} \wedge t^\bullet \subseteq P_{\text{prod}}\} \quad (3)$$

$$T_{\text{sell}} = \{t \in T \mid \bullet t \subseteq P_{\text{prod}} \wedge t^\bullet \subseteq P_{\text{mon}}\} \quad (4)$$

So transitions in T_{mon} and T_{prod} are between places in the money part; transitions in T_{prod} are between places in the production part. For instance, a T_{buy} transition consumes value from the money part and produces value in the production part. We use the partitioned transitions to filter the flow matrix' columns: $\mathbf{G}_x[i, j] = \mathbf{G}[i, j]$ if $j \in T_x$ and zero otherwise.

$$\mathbf{G} = \mathbf{G}_{\text{mon}} + \mathbf{G}_{\text{buy}} + \mathbf{G}_{\text{prod}} + \mathbf{G}_{\text{sell}} \quad (5)$$

If we also split matrix $\mathbf{G}$ into $\mathbf{B} - \mathbf{F}$ to see the difference between production and consumption, we can express connections between the value cycle's parts. Let $\mathbf{t}$ be a tour. Tour result $\mathbf{G} \cdot \mathbf{t}$ is factored into a causal chain as follows

$$(\mathbf{B}_{\text{mon}} + \mathbf{B}_{\text{buy}} + \mathbf{B}_{\text{prod}} + \mathbf{B}_{\text{sell}} - \mathbf{F}_{\text{mon}} - \mathbf{F}_{\text{buy}} - \mathbf{F}_{\text{prod}} - \mathbf{F}_{\text{sell}}) \cdot \mathbf{t} \quad (6)$$

The various terms in this expression are the input and output of the different steps in the causal chain.

Grouping the money and production parts generates the following equations. These equations are well known in auditing. Equation (7) is Kirchhof's law of preservation for the production process. It shows that in a tour the input and output of production cancel out. Equation (8) shows the value jump. It expresses the difference between the monetary values produced and consumed.

PROPOSITION 1.

$$(\mathbf{B}_{\text{buy}} + \mathbf{B}_{\text{prod}} - \mathbf{F}_{\text{prod}} - \mathbf{F}_{\text{sell}}) \cdot \mathbf{t} = 0 \quad (7)$$

$$(\mathbf{B}_{\text{sell}} + \mathbf{B}_{\text{mon}} - \mathbf{F}_{\text{mon}} - \mathbf{F}_{\text{buy}}) \cdot \mathbf{t} = \mathbf{G} \cdot \mathbf{t} \quad (8)$$

PROOF The matrix in $(\mathbf{B}_{\text{buy}} + \mathbf{B}_{\text{prod}} - \mathbf{F}_{\text{prod}} - \mathbf{F}_{\text{sell}}) \cdot \mathbf{t}$ is exactly the flow matrix filtered for the production places and does not contain the money place. Because $\mathbf{t}$ is a tour, the product must be zero according to Definition 4. The second equation then follows immediately because the tour result $\mathbf{G} \cdot \mathbf{t}$ must result from the other parts of Equation 6. $\square$

These algebraic notions are part of a theory developed by Griffioen [16]. They form a semantics for a programming language called Pacioli (<http://pgriffel.github.io/pacioli/>). Pacioli is a Domain Specific Language, specifically developed to capture the essence of the accounting domain. The advantage of Pacioli is that it allows to express transactions in their proper units of measurement. This is similar to a type system, as we find in many programming languages. Pacioli can also be used for comparing predictions to large amounts of real data. It is scalable.

5. CASE: CONTRACT MONITORING

The case is concerned with a legal dispute about a contract for public transport, granted to a public transport provider by a contract manager on behalf of a group of municipalities in a region. The governance setting of the case is similar to that of the SRE case [10]. Contract monitoring has been automated. Therefore, possible sources of evidence are all produced by information systems. In this case study we

reproduce the reasoning of a domain expert about appropriateness of this evidence.

Case description. Since 2011 a transport provider provides taxi-bus transport services for secondary school pupils who need to attend special schools. This type of transport is highly regulated. Regulations stipulate safety norms for the vehicle, qualifications of the driver etc. In the tender procedure the municipalities have laid down requirements concerning maximal waiting times, durations, routing, etc. The contract specifies the following revenue model. Based on trip requests (school children travel from A to B at time t , at week day w), the route planner calculates the daily routes at each time of day, and the number of children per route. Given the capacity of a taxi-bus, this gives the number of occupied vehicle kilometres (Dutch: beladen kilometers). Parties agreed that planning software ABC , which is accessible over the internet, is used to predict the expected route and duration. This prediction is based on standard speeds (100 km/h for motorways, 70 km/h for main roads, 40 km/h in town). For each route, this gives the occupied vehicle hours (Dutch: beladen uren), priced at a certain tariff: *revenue per route = price * occupied vehicle hours*.

The taxi-bus market is very competitive, with low margins and increasing costs. The transport provider owns and operates the vehicles. Handling travel requests and scheduling are delegated to different departments and are supported by software for CRM, driver scheduling, and route planning.

Parties explicitly agreed that the predicted, not the actual route duration can be charged. Nevertheless, according to the contract, the transport provider is required to provide to the contract manager evidence of the actual trips performed in each month, as part of the invoicing and payment process. The contract contains a data protocol specifying in detail which kinds of data must be included (route; times and durations; passenger details, etc.). On the basis of this data, the accountant of the municipalities can verify whether public money is spent according to the relevant regulations.

There is a special clause in the contract stating that in case of structural traffic congestion, measured per route over three months, a congestion surcharge can be awarded.

Issue. Apparently, there have been delays. The transport provider now claims a congestion surcharge. The municipalities want to find out if there have been delays, if such delays were caused by structural traffic congestion, and if they are not already covered by the contract price. It is defensible that congestion is part of the business risks of a transport provider and that no surcharge is due. So, is there any evidence that proves that the extra driving hours are actually caused by structural traffic congestion? Parties have hired a consultant in the role of domain expert in public transport services to investigate the issue.

Data collection. The researchers have access to the investigation report of the domain expert and the underlying data, and to descriptions of the processes by which it was derived. As the case is not yet settled, names have been shielded. The value nets (Figure 4 and Figure 5) were drawn by the researchers based on an analysis of the case material, and were validated by the domain expert.

Objective. The task for the domain expert was formulated as follows. "On the basis of the current registrations, is it possible to develop a systematic calculation to determine a congestion surcharge, as mentioned in the specifications,

which is sufficiently transparent and verifiable for both parties?” For the way to calculate the size of a possible surcharge no prior specifications were made.

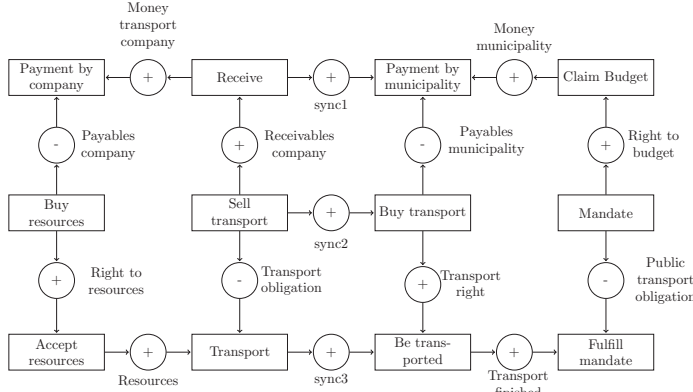


Figure 4: Interlinked value nets of transport provider and municipalities

Sources of Evidence. The domain expert assessed relevance and reliability of available data sources. We will use steps 1 – 7 of the Actor Analysis to structure the analysis. Each of these issues was indeed taken into account by the domain expert, although not necessarily in this order. Value nets and factored tours are used for conceptual analysis.

1. The actors are the transport provider and the contract manager, on behalf of municipalities. They have a countervailing interest. Figure 4 shows how their value nets are connected. Essentially, a municipality is buying and a transport provider is selling transport. The ‘sync’ places in the middle represent the sale. To deliver, the transport provider needs money to acquire resources like cars and drivers (left). Transport services are used by pupils (bottom), and in this way the municipality fulfils its social mandate (right). For this it is given a budget, which is used to pay (top).

2. The business of a transport provider classifies as a service of renting out capacity, namely transportation [33]. Its value net is shown in Figure 5. Planning is crucial for this type of service, shown on the bottom left of the value net. We can in principle cross-verify planned transport against actual transport (bottom right), and against what was paid.

3. Here is the main audit equation with the units listed: $revenue\ per\ route[€] = price[€] * occupied\ vehicle\ hours[h]$. Here we are interested in the difference between invoiced hours and actual hours: $difference = actual - invoiced$.

4. In general, a transport provider has reasons to overstate revenue(+), and therefore occupied vehicle hours(+). That is covered by reference to the independent *ABC* planning software. To claim a congestion surcharge, transport provider has reasons to overstate the difference(+), and therefore actual hours(+), or understate invoiced hours(-).

5. The monthly data file accompanying the invoice forms a trustworthy and objectively verifiable source of evidence, also for the actual hours. By contract, the transport provider was required every month to provide such data on the basis of its own systems (planning, GPS). The contract manager has verified the data as part of the monthly payment process. These data are therefore undisputed. For some routes, no reliable data is available, and for those routes the transport provider can make no claim. Those routes are scoped out. The transport provider has the burden of proof.

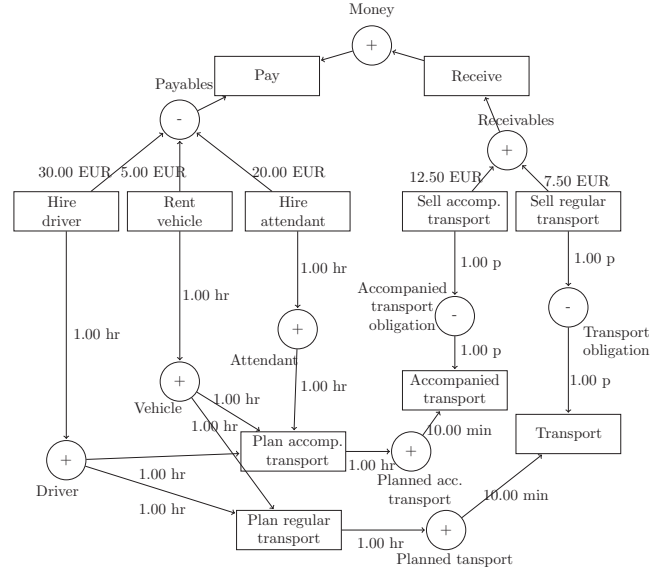


Figure 5: The value net of a transport provider

Next, all the planning and GPS data could be reconciled (matched per month) against the data files that underlie the actual hours. These sources are prepared under segregation of duties (planning department; tracking and tracing devices operated by taxi drivers). In addition, data for the invoiced hours were compared with historical data of the *ABC* software; no deviations were found.

So there are indeed appropriate sources of evidence, both relevant (right routes and right period of time) and reliable, as the data was already verified by the contract manager, and could be reconciled with independent sources.

Analysis. Table 1 contains a sample of the differences between actual occupied vehicle minutes and invoiced occupied vehicle minutes, for a selection of routes in the morning and afternoon. It is clear that there are large differences in the morning. Differences in the afternoon are smaller. Similar calculations are done for all Mondays, Tuesdays and so on.

The significance of the calculated differences per weekday was investigated. Statistically the populations per weekday are dependent, so a t-test for pairwise observations is applicable. It was first established that the data sets, separated

Morning					
route	occurrences	actual	invoiced	act-inv	% diff
32	57	81	65	16	24%
28	50	62	50	12	24%
51	58	61	53	8	15%
54	22	61	56	5	9%
50	56	32	30	2	4%
80	59	53	52	1	2%
Afternoon					
route	occurrences	actual	invoiced	act-inv	% diff
32	57	71	68	3	5%
28	58	47	43	4	9%
51	52	51	48	3	5%
54	19	56	54	2	3%
50	58	30	30	0	0%
80	59	49	48	1	2%

Table 1: Sample of findings for a selection of routes

for morning and afternoon, show a normal distribution. For the morning routes the zero-hypothesis could be rejected: the differences are significant. For the afternoon the differences are not significant. Can we infer that these differences can be attributed to traffic congestion so a surcharge is due?

The key question is whether we really understand the issue. Value nets are useful for conceptual analysis. Have another look at Figure 4. Typically municipalities care about transportation for school pupils, as this is how they acquire their budget. So although municipalities pay for the duration of planned routes, they want to account for the pupils served. When the value net is modelled as defined in Section 4, the factored tour (Table 2) reveals the composite unit that links pupils with delivered and invoiced transport, namely the capacity utilization factor. For instance 3600 s worth of planned transport, with a capacity utilization factor of 6, converts into 600 pupil seconds, i.e. 10 minutes. The capacity utilization factor needs to be distinguished from the capacity utilization degree. Taxi-buses have on average a maximum capacity of 8, so in this example the capacity utilization degree is 75%. Similarly, in Figure 5 1 hr of planned transport is converted into 6 * 10 minutes worth of transport per pupil (p).

The revenue model can now be written as: *revenue per route = price * number of pupils planned on that route * planned route duration per pupil*.

The transport provider claims that the actual trip duration per pupil is longer than the 10 minutes planned. Remember we know that there is a difference, at least in the morning, but we are not certain whether the differences are attributable to traffic congestion. Realise that routes are driven consecutively, so differences accumulate.

If we would have the traces (timestamps and GPS coordinates) of the vehicle registered from the first pick up till the last delivery at school, then we would be able to compute the differences between the planned schedule per pupil and actual performance. Actually, we should analyse all the traces of a vehicle so we can compute the distribution of differences over routes per pick-up and drop-off. It is to be expected that differences occur only on certain parts of the road-system. Such outcomes can then be compared with information about traffic jams in the area, which is commonly known and verifiable. This would strengthen the conviction, probably for both parties, that the extra time is attributable to traffic congestion. It would also strengthen the appropriateness of the audit evidence. A t-test shows that the morning differences per route are significant, but this does not provide a solid base for inferring that the differences are attributable to traffic jams.

In this case individual vehicle traces were not provided, so parties could not get reasonable assurance whether the differences are due to traffic congestion. Additional negotiations are necessary to settle the dispute.

Outcomes. First, it is possible to reach agreement on a method of how to calculate congestion. There is relevant and reliable evidence from the invoicing process.

Second, from the evidence presented as part of the monthly invoicing process, it appears there is a statistically significant difference between the invoiced hours, and the actual driven hours, but only for the mornings. The value net analysis indicates, that what matters are the planned route durations per pupil. Further analysis of individual vehicle traces would have to show whether this difference is structural for

some specific parts of the road network, and can therefore be attributed to traffic congestion. The current evidence is inconclusive in this respect.

Third, the contract price did not yet cover the business risk of congestion. So there is reason to award a congestion surcharge. The basis of calculating the surcharge can be derived from the value net analysis: the number of extra hours attributable to congestion. How these differences should be priced, however, is still being debated. Although parties have a legal dispute, they still prefer to settle out of court, as they would like to continue the working relationship.

6. CONCLUSIONS

When evidence is being generated by information systems, reasoning about legal evidence involves a design issue. Management must set up a reliable system of internal controls. It is the task of the auditor to evaluate the strength of the controls, and thereby the strength of evidence. Therefore it makes sense to look at evidence from the perspective of auditing theory. *How should auditors assess reliability of evidence that is being generated by information systems?*

In this paper we have tried to answer this question, by giving a background on auditing theory. We explained the notion of sufficient appropriate evidence, that involves the size of the sample, as well as the relevance and reliability of the evidence itself. We presented a practical actor analysis approach, to determine reasons for overstatement or understatement of accounts, and thus assess the relative reliability of possible sources of evidence. We have also given a brief overview of reasoning about evidence using models, highlighting the distinction between exceptions and violations.

Most computational approaches to compliance monitoring are based on process models like BPMN, that tend to focus on individual events and the order in which they occurred. For other domains, however, other representations of desirable behaviour are more appropriate. In particular, when disputes arise about contract enforcement, or about financial transactions, representations that capture the exchange of economic value are more appropriate. Typically, such disputes are about the effects of events on an aggregate level, not about individual transactions.

To assess the quality of evidence, we advocate use of model-based auditing, which uses audit equations to capture expected relations between the flow of money and goods or services. These equations form a value cycle model, which is specific to a type of business or sector. The model can be made mathematically precise using an algebraic theory that captures the notion of a value net. Value nets also correspond to existing notations and conceptions used in the accounting domain; therefore they can also be used and understood by auditors and domain experts. This is born out by our experience in teaching professionals. So value nets form a bridge between formal models, and accounting conceptions. The underlying algebra serves as a semantics for the Pacioli programming language, in which various tests and analyses can be programmed and analysed.

From the model, we derive expectations to be compared with actual data. When there is a difference, and it is significant, several assumptions may be challenged. The difference can be attributed to (a) applicability of the model, (b) application of the model, (c) coverage of the model (exceptions), or (d) reliability of the data. Only when all possible rebuttals are ruled out, can we establish (e) a material

Place	$F_{\text{buy}} \cdot t$	$B_{\text{buy}} \cdot t$	$C \cdot t$	$F_{\text{sell}} \cdot t$	$B_{\text{sell}} \cdot t$
payables	3500 c	-	-	-	-
vehicle	-	3600 s	-	-	-
driver	-	3600 s	-	-	-
planned reg. transport	-	-	3600 s	-	-
reg. transport obligation	-	-	-	6 p	-
receivables	-	-	-	-	4500 c

Place	$F_{\text{buy}} \cdot t$	$B_{\text{buy}} \cdot t$	$C \cdot t$	$F_{\text{sell}} \cdot t$	$B_{\text{sell}} \cdot t$
payables	5500 c	-	-	-	-
vehicle	-	3600 s	-	-	-
driver	-	3600 s	-	-	-
attendant	-	3600 s	-	-	-
planned acc. transport	-	-	3600 s	-	-
acc. transport obligation	-	-	-	6 p	-
receivables	-	-	-	-	7500 c

Table 2: Factored tour for the transport case. Columns show the chain of events

misstatement (violation). It appears that reference to an independently motivated model is a typical reasoning pattern used by domain experts. Further research should elaborate the precise argumentative structure of these kinds of claims and rebuttals. We expect that this type of reasoning about the use of models, could be captured by an argumentation scheme with corresponding critical questions [35], similar to the scheme proposed by [8].

To provide an initial evaluation of the usefulness of the approach, we have looked at a case study of a dispute over a contract for public transport. The transport provider claims a surcharge, because of structural congestion. The auditor, as domain expert, must investigate the evidence. The case shows that it matters that there is a mutually agreed revenue model (a), and an agreed procedure (b) for calculating routes (*ABC* software). Moreover, the data is considered reliable, since it was verified by both parties in the invoicing process (c). There is no reason to assume violations of the contract that could explain the differences, in particular because the transport provider is the claimant in this case (e). So all possible rebuttals (a, b, c, e) about the model, the data and underlying assumptions are countered. The remaining possibility is that the differences are explained by an exception to the model (d). In this case a plausible exception is traffic congestion. In other words, the domain expert established the presence of reliable data (countervailing interests), which counts as evidence of propositions or events (difference between planned and actual occupied vehicle hours), to show that a condition holds in the context of a legal dispute (structural congestion). That means that we can say that the approach is useful in principle; it helps to settle the crucial issues in the application domain.

In the case, we find two of the three styles of reasoning about evidence: (i) statistics to evaluate the strength of evidence, and (ii) argumentation to challenge and defend assumptions about the system and process by which evidence is generated. The narrative style (iii) is not found in the case, although the notion of a tour is related to a narrative; both represent a causal chain of events.

Our focus on auditing and on transactions as the central notion for characterizing evidence of compliance, also has drawbacks. In particular, it restricts the notion of compliance to properties that can be expressed in terms of value exchanges. So like BPMN process models, the account is

not fully generalizable. Our experience in teaching shows however, that accounting notions of stock keeping and Kirchhoff's law of preservation can also be fruitfully applied to business process modelling at an aggregate level. Consider service requests at a help desk. A set of requests forms a kind of inventory. The work-flow principle that all requests must eventually be either granted or denied with an explanation, and that no request may get lost, is an instance of the preservation axiom. Also for such cases we can develop a value net and derive useful audit equations.

7. REFERENCES

- [1] American Accounting Association. Report of the committee on basic auditing concepts. *The Accounting Review*, XLVII:14–74, 1972.
- [2] T. J. Anderson, D. A. Schum, and W. L. Twining. *Analysis of Evidence*. Cambridge University Press, Cambridge, 2005.
- [3] F. Bex, P. van Koppen, H. Prakken, and B. Verheij. A hybrid formal theory of arguments, stories and criminal evidence. *Artificial Intelligence and Law*, 18:123–152, 2010.
- [4] F. Bex, H. Prakken, C. Reed, and D. Walton. Towards a formal account of reasoning about evidence: argumentation schemes and generalisations. *Artificial Intelligence and Law*, 11:125–165, 2003.
- [5] J. Black. Regulatory conversations. *Journal of Law and Society*, 29(1):163–196, 2002.
- [6] J. Blokdijk. Tests of control in the audit risk model: effective? efficient? *International Journal of Auditing*, 8:185–194, 2004.
- [7] J. Blokdijk, F. Drieënhuizen, and P. Wallage. *Reflections on auditing theory, a contribution from the Netherlands*. Limperg Instituut, Amsterdam, 1995.
- [8] B. Burgemeestre, J. Hulstijn, and Y.-H. Tan. Value-based argumentation for justifying compliance. *Artificial Intelligence and Law*, 19(2-3):149–186, 2011.
- [9] B. Burgemeestre, J. Hulstijn, and Y.-H. Tan. Value-based argumentation for designing and auditing security measures. *Ethics and Information Technology*, 15:153–171, 2013.
- [10] R. Christiaanse and J. Hulstijn. Control automation to reduce costs of control. *International Journal of*

Information System Modeling and Design, 4(4): 27–47, 2013.

- [11] COSO. Internal control - integrated framework. Committee of Sponsoring Organizations of the Treadway Commission, 1992.
- [12] J. L. Dietz. The deep structure of business processes. *Communications of the ACM*, 49(5):59–64, 2006.
- [13] P. Elsas. *Computational auditing*. PhD thesis, Vrije Universiteit, Amsterdam, 1996.
- [14] J. Gordijn and J. M. Akkermans. Value-based requirements engineering: Exploring innovative e-commerce ideas. *Requirements Engineering*, 8(2):114–134, 2003.
- [15] G. Governatori and S. Sadiq. *The journey to business process compliance*, pages 426–445. IGI Global, 2009.
- [16] P. Griffioen. Type inference for linear algebra with units of measurement. Technical report, Centrum Wiskunde en Informatica (CWI), 2013.
- [17] P. Griffioen, P. Elsas, and R. van de Riet. *Analyzing Enterprises: the value-cycle approach*, pages 685–697. Springer Verlag, Berlin, 2000.
- [18] IFAC. ISA 530: Audit sampling. International Auditing and Assurance Standards Board (IAASB), International Federation of Accountants (IFAC), 2008.
- [19] IFAC. ISA 500: Audit evidence. International Auditing and Assurance Standards Board (IAASB), International Federation of Accountants (IFAC), 2012.
- [20] Y. Ijiri. *The foundations of accounting measurement: a mathematical, economic, and behavioral inquiry*. Prentice-Hall, 1967.
- [21] B. Klievink, E. van Stijn, D. Hesketh, H. Aldewereld, S. Overbeek, F. Heijmann, and Y.-H. Tan. Enhancing visibility in international supply chains: The data pipeline concept. *International Journal of Electronic Government Research*, 8(4):14–33, 2012.
- [22] W. Knechel, S. Salterio, and B. Ballou. *Auditing: Assurance and Risk*. Thomson Learning, Cincinnati, 3rd edition, 2007.
- [23] R. B. Korobkin. Behavioral analysis and legal form: Rules vs. principles revisited. *Oregon Law Review*, 79(1):23–60, 2000.
- [24] L. Lindahl and J. Odelstad. Intermediaries and intervenients in normative systems. *Journal of Applied Logic*, 6(2):229–250, 2008.
- [25] L. A. Maines and J. M. Wahlen. The nature of accounting information reliability: Inferences from archival and experimental research. *Accounting Horizons*, 20(4):399–425, 2006.
- [26] J.-J. Meyer and R. Wieringa. *Deontic Logic in Computer Science: Normative System Specification*. John Wiley and Sons, 1993.
- [27] J. Pearl. Fusion, propagation, and structuring in belief networks. *Artificial Intelligence*, 29(3):241–288, 1986.
- [28] M. Power. The risk management of nothing. *Accounting, Organizations and Society*, 34:849–855, 2009.
- [29] H. Prakken, D. Ionita, and R. Wieringa. Risk assessment as an argumentation game. In J. Leite, editor, *14th International Workshop on Computational Logic in Multi-Agent Systems (CLIMA XIV)*, pages 357–373. Springer Verlag, 2013.
- [30] R. Reiter. A theory of diagnosis from first principles. *Artificial Intelligence*, 32(1):57–95, 1987.
- [31] A. Rozinat and W. M. P. van der Aalst. Conformance checking of processes based on monitoring real behavior. *Information Systems*, 33(1):64–95, 2008.
- [32] J. R. Searle. *The Construction of Social Reality*. The Free Press, 1995.
- [33] R. W. Starreveld, B. de Mare, and E. Joels. *Bestuurlijke Informatieverzorging (in Dutch)*, volume 1. Samsom, Alphen aan den Rijn, 1994.
- [34] W. Wagenaar, P. v. Koppen, and H. N. Y. Crombag. *Anchored Narratives: The Psychology of Criminal Evidence*. St. Martin’s Press, New York, 1993.
- [35] D. N. Walton. *Argument Schemes for Presumptive Reasoning*. Lawrence Erlbaum Associates, Mahwah, 1996.
- [36] D. N. Walton and E. C. Krabbe. *Commitment in Dialogue: Basic Concepts of Interpersonal Reasoning*. State University of New York Press, 1995.
- [37] H. Weigand and P. Elsas. Model-based auditing using REA. *International Journal of Accounting Information Systems*, 13:287–310, 2012.
- [38] P. Westerman. Who is regulating the self? Self-regulation as outsourced rule-making, In: M. Hertogh and P. Westerman, editors, *Self-Regulation and the Future of the Regulatory State*, pages 23–36, University of Groningen, 2009.
- [39] J. Wigmore. *The Principles of Judicial Proof or the Process of Proof as Given by Logic, Psychology, and General Experience and Illustrated in Judicial Trials*. Little, Brown and Company, Boston, 2nd edition, 1931.